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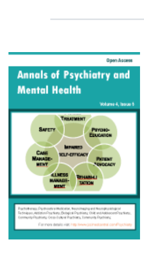
Micronutrient Treatment of Emotional Dyscontrol Following Traumatic Brain Injury

Emotional dyscontrol following traumatic brain injury (TBI) impairs social relationships and employability. Micronutrients (minerals, vitamins) stabilize emotional lability in psychiatric patients, and various individual nutrients have been used to treat experimental brain injury in laboratory animals in the acute phase. However, the current case report appears to be the first documentation of micronutrients resulting in normalization of emotion regulation in a long-standing brain injury in a human.

Case presentation: A broad-spectrum formula of micronutrients† was evaluated in a 35-year-old male who had incurred a severe TBI eight years previously. Resolution of most post-TBI symptoms was achieved during those eight years, but not his episodic loss of emotional control, which psychiatrists evaluated as being permanent. The trial of micronutrients began after five weeks of baseline symptom monitoring with a mood stability scale. By three months mood stability had improved markedly according to data submitted by two raters (the patient and his clinician) who were blind to each other's evaluations. Data collection continued for one year, showing significant improvement (p<.0001), at which time the patient reported that his emotional control had returned to his pre-TBI level. The improvements led to his establishing his own business and improving his family relationships.

Micronutrient treatment resulted in resolution of this patient's longstanding post-TBI emotional dyscontrol. Broad-spectrum micronutrient formulas are showing benefit for the treatment of mood lability in various types of psychiatric patients; this report indicates there is also potential value in using them for the emotional dyscontrol found in post-TBI patients.



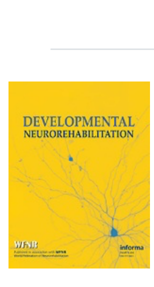
Micronutrient Treatment of Emotional Dyscontrol Following Traumatic Brain Injury

Kaplan, B. J., Leaney, C., Tsatsko, E. Annals of Psychiatry and Mental Health. 29 August 2016.

FULL TEXT

Micronutrient formulation greatly improves brain function in rats.

Researchers at the world-renowned Canadian Centre for Behavioural Neuroscience conducted a series of controlled studies with rats fed a micronutrient formulation co-developed† by Hardy Nutritionals® founder David Hardy to test its effects on brain injuries and brain-related disorders. Neuroscientists performed surgery on a group of young rats, removing either the entire frontal lobe of the brain or part of the parietal lobe. The researchers then fed one group of these rats Purina Rat Chow (considered to be an ideally fortified lab rat diet with 22 added vitamins and minerals) and another group the same base diet fortified with the more complete, specially-processed micronutrient formulation. Brain recovery was significantly enhanced in the micronutrient-fed rats. As the rats recovered, researchers observed that the micronutrient-fed rats showed significantly calmer behavior and dramatically improved cognitive function compared with Purina-fed rats. The brain cells of micronutrient rats were also more complexly-branched and better connected with neighboring cells than the brain cells of Purina-fed rats. The calmer behavior and enhanced cognitive function of micronutrient-treated rats are consistent with results documented by researchers around the world in many people with mood, anxiety, and behavioral disorders. The enhanced brain cell connectivity observed by researchers supports the use of micronutrient formulations for human brain recovery in conjunction with brain rehabilitation therapies.



Factors influencing frontal cortex development and recovery from early frontal injury.

Halliwell C, Comeau W, Gibb R, Frost DO, Kolb B. Developmental Neurorehabilitation. 2009;12(5):269-78.

FULL TEXT

†The micronutrient formulation studied was a pre-2013 version of *Truehope EMPowerplus* which was co-formulated by David Hardy and Anthony Stephan. *Truehope EMPowerplus* is a registered trademark of The Synergy Group of Canada Inc., which was co-founded by David Hardy and Anthony Stephan in 1999. David Hardy officially resigned as a shareholder of The Synergy Group of Canada Inc. and director of Truehope Nutritional Support Ltd. in 2013 to focus his efforts exclusively on Hardy Nutritionals®.

STUDIES MENU

- 1-year naturalistic follow-up of a Randomised Double-Blind, Placebo-Controlled Trial ("NoMAD") Exploring the Effectiveness of Micronutrients in Improving Symptoms of Anxiety and Depression
- Gut microbiome changes with micronutrient supplementation in children with attention-deficit/hyperactivity disorder: the MADDY study
- Potential link between copy number variation and abnormal genome wide DNA methylation profile in an individual with severe ADHD and a strong response to micronutrient treatment
- Investigation of a broad-spectrum micronutrient formulation as a possible precipitant of pharmacokinetic micronutrient-drug interactions
- Broad-Spectrum Micronutrients or Antidepressants for Antenatal Depression: Effect on Maternal and Infant Birth Outcomes in an Observational Secondary Analysis of NUTRIMUM
- Efficacy and safety of a mineral and vitamin treatment on symptoms of antenatal depression: 12-week fully blinded randomised placebo-controlled trial (NUTRIMUM)
- Problems most concerning to parents of children with ADHD and emotional dysregulation in a randomized controlled trial of multinutrients: MADDY secondary analysis
- Treatment response to supplemental nutrients for ADHD is independent of diet quality: the MADDY Study RCT
- Micronutrients for ADHD in youth (MADDY) study: comparison of results from RCT and open label extension.
- Exploring the impact of antenatal micronutrients used as a treatment for maternal depression on infant temperament in the first year of life
- Household Food Insecurity Is Associated with Symptoms of Emotional Dysregulation in Children with Attention Deficit Hyperactivity Disorder: The MADDY Study.
- Effect of antenatal micronutrient or antidepressant exposure on Brazelton neonatal behavioral assessment scale (NBAS) performance within one-month of birth
- Efficacy and safety of a vitamin-mineral intervention for symptoms of anxiety and depression in adults: A randomised placebo-controlled trial "NoMAD"
- Non-pharmacological interventions for attention-deficit hyperactivity disorder in children and adolescents
- Inflammation and Vitamin C in Women with Prenatal Depression and Anxiety: Effect of Multinutrient Supplementation.
- Diet Quality as a Moderator of Response to Multinutrients for ADHD and Emotional Dysregulation: The MADDY RCT.
- Micronutrients for Attention-Deficit/Hyperactivity Disorder in Youth: A Placebo-Controlled Randomized Clinical Trial
- Paediatric adverse event rating scale: a measure of safety or efficacy? Novel analysis from the MADDY study
- Clinician guidelines for the treatment of psychiatric disorders with nutraceuticals and phytoceuticals: The World Federation of Societies of Biological Psychiatry (WFSBP) and Canadian Network for Mood and Anxiety Treatments (CANMAT) Taskforce
- Parents' priorities and preferences for treatment of children with ADHD: Qualitative inquiry in the MADDY study
- Do Changes in Blood Nutrient Levels Mediate Treatment Response in Children and Adults With ADHD Consuming a Vitamin-Mineral Supplement?
- Multinutrients for the Treatment of Psychiatric Symptoms in Clinical Samples: A Systematic Review and Meta-Analysis of Randomized Controlled Trials
- A Holistic Self-learning Approach for Young Adult Depression and Anxiety Compared to Medication-Based Treatment-As-Usual
- Resting-state networks and neurometabolites in children with ADHD after 10 weeks of treatment with micronutrients: results of a randomised placebo-controlled trial
- Novel Mineral–Vitamin Treatment for Reduction in Cigarette Smoking: A Fully Blinded Randomized Placebo-Controlled Trial
- Human gut microbiome changes during a 10 week Randomised Control Trial for micronutrient supplementation in children with attention deficit hyperactivity disorder
- Mineral-Vitamin Treatment Associated with Remission in Attention-Deficit/Hyperactivity Disorder Symptoms and Related Problems: 1-Year Naturalistic Outcomes of a 10-Week Randomized Placebo-Controlled Trial
- Multinutrients for ADHD Youth (MADDY) Trial: Preliminary Trends for Treatment
- An Observational Preliminary Study on the Safety of Long-Term Consumption of Micronutrients for the Treatment of Psychiatric Symptoms
- Can we predict treatment response in children with ADHD to a vitamin-mineral supplement? An investigation into pre-treatment nutrient serum levels, MTHFR status, clinical correlates and demographic variables
- Methylomic changes in response to micronutrient supplementation and MTHFR genotype.
- Hospitalization cost of conventional psychiatric care compared to broad-spectrum micronutrient treatment: literature review and case study of adult psychosis
- Vitamin-mineral treatment improves aggression and emotional regulation in children with ADHD: a fully blinded, randomized, placebo-controlled trial.
- Vitamin–Mineral Treatment of ADHD in Adults: A 1-Year Naturalistic Follow-Up of a Randomized Controlled Trial
- Micronutrient Treatment of Emotional Dyscontrol Following Traumatic Brain Injury
- Effect of Micronutrients on Insomnia in Adults: A Multiple-Baseline Study
- A randomised trial of nutrient supplements to minimise psychological stress after a natural disaster
- Clinically Significant Symptom Reduction in Children with Attention-Deficit/Hyperactivity Disorder Treated with Micronutrients: An Open-Label Reversal Design Study
- Vitamin–mineral treatment of attention-deficit hyperactivity disorder in adults: double-blind randomised placebo-controlled trial.
- Moderators of treatment response in adults with ADHD treated with a vitamin-mineral supplement
- Psychological functioning 1 year after a brief intervention using micronutrients to treat stress and anxiety related to the 2011 Christchurch earthquakes: a naturalistic follow-up
- Nutritional and Safety Outcomes from an Open-Label Micronutrient Intervention for Pediatric Bipolar Spectrum Disorders
- Could yeast infections impair recovery from mental illness? A case study using micronutrients and olive leaf extract for the treatment of ADHD and depression.
- Use of micronutrients attenuates cannabis and nicotine abuse as evidenced from a reversal design: a case study.
- Efficacy and cost of micronutrient treatment of childhood psychosis.
- Micronutrient formulation outperforms Berocca Performance in reducing depression, anxiety, and stress in 91 earthquake survivors.
- Feasibility of a nutritional supplement as treatment for pediatric bipolar spectrum disorders.
- Safety and tolerability of Micronutrient formulations.
- Can micronutrients improve neurocognitive functioning in adults with ADHD and severe mood dysregulation? A pilot study.
- Micronutrient formulation reduces depression, anxiety, and stress in 17 earthquake survivors with ADHD.
- Micronutrients reduce stress and anxiety following a 7.1 earthquake in adults with Attention-Deficit/Hyperactivity Disorder
- Effect of micronutrients on behaviour and mood in adults with ADHD: Evidence from an 8-week open label trial with natural extension.
- Database analysis: Micronutrient formulation greatly improves bipolar and ADHD symptoms in 161 children and adolescents.
- Successful treatment of bipolar disorder II and ADHD with a micronutrient formula: a case study.
- Micronutrient formulation outperforms medications in 88 patients with autism.
- Micronutrient formulation greatly improves brain function in rats.
- Database analysis: Micronutrient formulation greatly improves bipolar symptoms in 358 adults.
- Successful treatment of OCD with a micronutrient formula following partial response to Cognitive Behavioral Therapy (CBT): a case study.
- Multinutrient supplement as treatment: literature review and case report of a 12-year-old boy with bipolar disorder.
- Micronutrient formulation greatly improves mood, behavior, and anxiety disorders in 9 children.
- Treatment of mood lability and explosive rage with minerals and vitamins: two case studies in children.
- Micronutrient formulation greatly improves bipolar symptoms in 19 adults.
- Micronutrient formulation greatly improves bipolar symptoms in 23 adults, adolescents, and children.
- Micronutrient formulation stabilizes mood in 11 bipolar adults.

★ REVIEWS

Consult with your doctor when altering your treatment regimen and before adjusting your medication dosing.

Information from Hardy Nutritionals® is for educational purposes only, and suggestions should not be considered medical advice.

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